

GKS2027_MedicalPhysics_Simulation_AKA_Nina

February 22, 2026

```
[1]: print("Objectif GKS 2027 : Physique Médicale")
```

Objectif GKS 2027 : Physique Médicale

```
[2]: # Loi de l'inverse du carré de la distance
D0 = 100 # Dose à 1 mètre
d = 2    # Nouvelle distance (2 mètres)
D1 = D0 / (d**2)
print(f"La dose à {d} mètres est de : {D1} mSv")
```

La dose à 2 mètres est de : 25.0 mSv

```
[3]: # 1. Installation de la bibliothèque si nécessaire
!pip install pydicom matplotlib

# 2. Importation des outils
import pydicom
from pydicom.data import get_testdata_files
import matplotlib.pyplot as plt

# 3. Récupération d'une image scanner (CT) de test gratuite
filename = get_testdata_files("CT_small.dcm")[0]
ds = pydicom.dcmread(filename)

# 4. Affichage des informations de physique médicale (Métadonnées)
print(f"Modalité : {ds.Modality}")
print(f"Épaisseur de coupe : {ds.SliceThickness} mm")
print(f"Tension du tube (KVp) : {ds.KVP} kV")

# 5. Affichage de l'image sur votre écran Haier
plt.imshow(ds.pixel_array, cmap=plt.cm.bone)
plt.title("Exemple de Scanner (CT) - Test GKS 2027")
plt.show()
```

Defaulting to user installation because normal site-packages is not writeable

Looking in links: /usr/share/pip-wheels

Collecting pydicom

Downloading pydicom-3.0.1-py3-none-any.whl.metadata (9.4 kB)

```
Requirement already satisfied: matplotlib in
/opt/conda/envs/anaconda-2025.12-py312/lib/python3.12/site-packages (3.10.6)
Requirement already satisfied: contourpy>=1.0.1 in
/opt/conda/envs/anaconda-2025.12-py312/lib/python3.12/site-packages (from
matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in
/opt/conda/envs/anaconda-2025.12-py312/lib/python3.12/site-packages (from
matplotlib) (0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in
/opt/conda/envs/anaconda-2025.12-py312/lib/python3.12/site-packages (from
matplotlib) (4.60.1)
Requirement already satisfied: kiwisolver>=1.3.1 in
/opt/conda/envs/anaconda-2025.12-py312/lib/python3.12/site-packages (from
matplotlib) (1.4.9)
Requirement already satisfied: numpy>=1.23 in
/opt/conda/envs/anaconda-2025.12-py312/lib/python3.12/site-packages (from
matplotlib) (1.26.4)
Requirement already satisfied: packaging>=20.0 in
/opt/conda/envs/anaconda-2025.12-py312/lib/python3.12/site-packages (from
matplotlib) (25.0)
Requirement already satisfied: pillow>=8 in
/opt/conda/envs/anaconda-2025.12-py312/lib/python3.12/site-packages (from
matplotlib) (12.0.0)
Requirement already satisfied: pyparsing>=2.3.1 in
/opt/conda/envs/anaconda-2025.12-py312/lib/python3.12/site-packages (from
matplotlib) (3.2.5)
Requirement already satisfied: python-dateutil>=2.7 in
/opt/conda/envs/anaconda-2025.12-py312/lib/python3.12/site-packages (from
matplotlib) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in
/opt/conda/envs/anaconda-2025.12-py312/lib/python3.12/site-packages (from
python-dateutil>=2.7->matplotlib) (1.17.0)
Downloading pydicom-3.0.1-py3-none-any.whl (2.4 MB)
      2.4/2.4
```

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Installing collected packages: pydicom

WARNING: The script pydicom is installed in

'/home/c97cd718-c243-4c66-9692-d316421a62a9/.local/bin' which is not on PATH.

Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.

Successfully installed pydicom-3.0.1

```
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ModuleNotFoundError
Cell In[3], line 5
```

```
Traceback (most recent call last)
```

```

2 get_ipython().system('pip install pydicom matplotlib')
4 # 2. Importation des outils
----> 5 import pydicom
6 from pydicom.data import get_testdata_files
7 import matplotlib.pyplot as plt

```

ModuleNotFoundError: No module named 'pydicom'

```

[4]: # 1. Installation de la bibliothèque si nécessaire
'pip install pydicom matplotlib

# 2. Importation des outils
import pydicom
from pydicom.data import get_testdata_files
import matplotlib.pyplot as plt

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filename = get_testdata_files("CT_small.dcm")[0]
ds = pydicom.dcmread(filename)

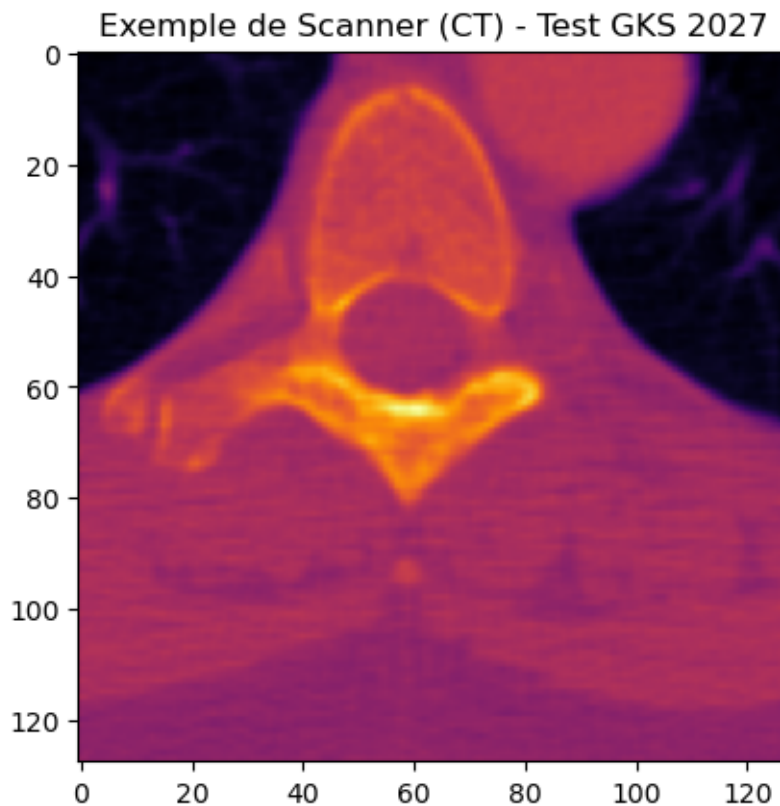
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print(f"Modalité : {ds.Modality}")
print(f"Epaisseur de coupe : {ds.SliceThickness} mm")
print(f"Tension du tube (KVp) : {ds.KVP} kV")

# 5. Affichage de l'image sur votre écran Haier
plt.imshow(ds.pixel_array, cmap='inferno')
plt.title("Exemple de Scanner (CT) - Test GKS 2027")
plt.show()

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python-dateutil>=2.7->matplotlib) (1.17.0)
Modalité : CT
Epaisseur de coupe : 5.000000 mm
Tension du tube (KVp) : 120 kV



[]: